

Requirement Specification:

TRUST Tech Deliverable 4a: Enable user GitLab & Docker images

Amit Karpe, Lead Engineer

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Revision History NA**Prepared By**

Name	Project Role	Signature	Date
Amit Karpe	Lead Engineer		31 Jan 2025

Reviewed By

Name	Project Role	Signature	Date
Yeo Zhen Xuan	Tech Lead		31 Jan 2025

Approved By

Name	Project Role	Signature	Date
Luke Lim	Synapxe Project Manager		31 Jan 2025
Kevin Lam	TRUST Business Lead		31 Jan 2025

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0.1 1. Introduction

0.1.1 1.1 References

- GovTech GCC Security Guidelines
- AWS Best Practices for Multi-Account Strategy

0.1.2 1.2 Abstract

This document defines the requirement specifications for deploying code, container repositories, and related cloud infrastructure in a non-internet environment (Singapore GCC) for Lifebit CloudOS, focusing on enabling private GitLab and AWS ECR usage.

0.2 2. Application Overview

0.2.1 2.1 Layers or architectural framework

The solution follows a multi-account AWS architecture for hosting CloudOS and related infrastructure components:

- AWS Service Account: Hosts CloudOS and GitLab Server for private code repositories.
- AWS Workload Account: Hosts AWS Batch and AWS ECR for containerized workload execution.

0.2.2 2.2 Deployment View

- GitLab Server: Deployed in the AWS Service Account, configured for private access.
- AWS ECR: Deployed in the AWS Workload Account, accessible only within AWS GCC accounts.
- AWS Batch: Utilized for running containerized jobs, with required IAM permissions.

0.2.3 2.3 Development View

- GitLab provides private repository hosting for researchers and engineers.
- CI/CD workflows will be managed within GitLab to integrate with AWS Batch.
- AWS ECR will store and manage private container images for workloads.

0.2.4 2.4 Architectural Mechanisms

- Secure networking between AWS accounts using VPC peering or AWS PrivateLink.
- IAM roles and policies restricting access to only necessary resources.

0.3 3. Architecturally significant requirements

- Deployment in a non-internet environment within Singapore GCC.
- Secure and private access for all services.
- Multi-account strategy for isolation and compliance.

0.4 4. Security

0.4.1 4.1 System Security

- Network access controls enforced via Security Groups and VPC.
- AWS KMS for encrypting sensitive data and images.

0.4.2 4.2 Data Security

- No external data access permitted; all data resides within AWS GCC.
- Private container images stored in AWS ECR with strict access policies.

0.4.3 4.3 Application Security

- GitLab access restricted to authorized users only.
- IAM roles configured for least privilege access.

0.5 5. Common Modules

- NA

0.6 6. Functional Modules

- NA

0.7 7. Message / Service Interface

- NA

0.8 8. Database Design

- NA

0.9 Appendix A

- NA